

Alternatives for Financing Social Security in Luxembourg by Resident and Cross-Border Households

Philippe Liégeois^{1*}

¹Luxembourg Institute of Socio-Economic Research/LISER, 4366 Belval Esch-sur-Alzette, Luxembourg

Abstract Given the rapidly evolving structural socio-demographic determinants in Luxembourg (ageing, migrations, cross-border households) and social needs induced, the concern about the funding of the Luxembourg social security system is high on the agenda. This concern could become even more pressing over time, given Luxembourg's specificity in several respects. According to the 2024 Ageing Report of the EPC's Ageing Working Group, Luxembourg might face an increase of age-related expenditure from 17.2% of GDP in 2022 to 27.9% in 2070, mostly due to pensions (+ 8.3% of GDP over the period). This is the biggest increase expected in EU-countries, a first particularity for Luxembourg. A second specificity is the importance of cross-border commuters in the Luxembourg economy. These represent 43% of total employment in 2022. Given such a context, the country's social players are looking for avenues of reflection for future concrete proposals. This paper precisely aims to contribute to such a debate. It examines day-after impact of hypothetical parametric changes in social contributions and personal income taxes (the "alternatives") on the distribution of household disposable income and total public financial receipts from these sources for Luxembourg. Moreover and given the importance of cross-border commuters for the country, we need a microsimulation modelling –EUROMOD-based– covering both residents and cross-border commuters' households, the latter population involving an essential innovative extension to previous assessments. We emphasize the structural discrepancies between residents and cross-border households in terms of socio-economic status as well as regarding gross labor and taxable income. Consequently, we show that total receipts from residents are greater than those from cross-border households, even when controlling for population size. Next, we examine 42 alternatives based on the concerns of a key Luxembourg social partner in the context of an ongoing public debate. This examination takes into account the values achieved for a triplet of standard indicators chosen for their simplicity and acceptability to a broad public, and this within the framework of an independent external expertise: total revenues (cross-border households included), the inequality Gini coefficient and the poverty rate (the latter two for the resident population only). We then complete our detailed overview with an evaluation of each alternative according to the selected dimensions altogether. Finally, we evoke a basic "Global Performance Index" which may enlighten conclusions derived from a one or two-dimensional analysis. Although the analysis is specific to Luxembourg, the policy alternatives and methodology considered here may also be relevant for other countries with comparable socio-economic fundamentals, particularly in the context of the EU-wide concern regarding the fiscal implications of population ageing.

JEL classification: C63, D31, D78, H24, H55, P43, P41

DOI: <https://doi.org/10.34196/ijm.00340>

***For correspondence:**
Philippe.Liegeois-ext@liser.lu

©This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Author Keywords: static microsimulation, cross-border households, EUROMOD, HFCS data, Comparison of socio-fiscal policies, Luxembourg
© 2026, Liégeois.

1. Motivation and Context of the Paper

Given the rapid evolution of structural socio-demographic determinants in Luxembourg (ageing, migration, cross-border households) and the resulting social needs (pensions, healthcare, dependency), the question of how to finance Luxembourg's social security system is high on the agenda and could become even more pressing over time.

In its baseline scenario, the 2024 Ageing Report from the EU-Economic Policy Committee's Ageing Working Group is expecting for Luxembourg an increase of age-related expenditure from 17.2% of GDP in 2022 to 27.9% in 2070, mostly due to pensions (+ 8.3% of GDP over the period).¹ This is the biggest increase expected in EU-countries, a first particularity for Luxembourg.

A second specificity is the importance of cross-border commuters in the Luxembourg economy. Those do matter by large, representing more than 40% of total employment (41% in 2017, 43% in 2022), hence significantly participating in the funding of social security, for example through social contributions and personal income taxes.²

Given such a situation, the country's social players are looking for avenues of reflection for future concrete proposals. In particular, the *Chambre des Salariés du Luxembourg* (CSL), a key social partner, has initiated a study aimed at launching a debate on the impact of hypothetical changes in social contributions and personal income tax (hereafter the "alternatives").³ This quest is formulated by highlighting four aspects.

Firstly, the main objective of the study is to examine funding possibilities. We do not consider other relevant issues related to public spending. Furthermore, given the nature of the Luxembourg economy, we should involve both residents and cross-border households.

Secondly, despite this keen interest in funding possibilities, distributive aspects are also at stake. What might be the impact of alternatives on income inequality, poverty, well-being, etc.? However, cross-border households are only partially covered by the microdata at our disposal and are embedded in a broader population that is not exclusively made up of commuters. Consequently, this examination of distributions for non-residents, while possible, can only be partial. The present analysis will therefore be limited mainly to residents in this respect, which is in any case relevant from a political point of view and an important first step towards a more comprehensive view.

Thirdly, the alternatives discussed here are the result of internal debates at CSL, which is responding to the concerns of its main affiliates as part of an ongoing public debate. The changes envisaged are parametric (e.g. thresholds, tax rates and social security contributions). We are therefore not concerned here with non-parametric reforms which, for example, would change the nature of the tax bases considered for calculating social security contributions or other taxes such as those based on consumption.

Finally, the analysis must be as transparent and independent as possible, hence a request to the Luxembourg Institute of Socio-Economic Research (LISER) to carry out a preliminary analysis. Dedicated tools can be found in the public administration in general or in other relevant bodies, but they are often either not very accessible to the social partners, unclear as to the assumptions made or the content, incomplete, or simply considered too much driven from external spheres with concerns differing from CSL's ones. Whether a reality or an *a priori* fear, such an interpretation is quite common in public debate, and can be partially addressed by independent analysis and the use of tools that are as simple and interpretable as possible.⁴

This is why we have chosen to build the present analysis on a pair of EUROMOD-type static microsimulation models, one for the resident population, the other targeting cross-border households, the latter population involving an essential innovative extension to previous assessments.⁵ Microsimulation techniques are well suited to generate missing information (e.g. individual taxes and benefits, if unknown from other sources) and analyze the distributional impact of changes in the socio-demographic environment (including the search for winners and losers). Such a technique is particu-

1. European Commission – **European Commission – DG ECFIN (2024)**.

2. **STATEC (2019)** for 2017 - **STATEC (2023)** for 2022.

3. <https://www.csl.lu/en/>

4. See **Blond-Hanten and Thomas (2014)** for a larger view.

5. CSL – LISER project « *Alternative Ways for Funding the Luxembourgish social security system, with distributional effects* » (2023a and 2023b). The full reports by (**Liégeois, 2023a; Liégeois, 2023b**) are for internal use only but **Liégeois (2025a)** is grounding on them and developing some modelling aspects to be made accessible to a larger audience.

larly relevant when the interactions induced by changes are non-linear, such as those resulting from the application of complex socio-fiscal policies.

More specifically, the EUROMOD platform is also well established, rapidly applicable, considered reliable and simple enough to analyze the background interactions leading to overall results. EUROMOD allows us to take a representative snapshot of a population for a given year and apply several policy rules to this population (calculating social contributions, benefits, income tax, etc. for each household or individual), leading to a distributional view of the “welfare” resulting from this specific social and fiscal policy. If we modify the rules applied, we obtain an alternative distribution, and we can focus both on overall changes (in income and expenditure) and on sub-groups particularly affected, positively or negatively, by this alternative.

The simple “day after” EUROMOD approach is therefore seen here by the CSL as an essential step towards the identification of avenues that could be of interest and explored in greater depth at a later stage when examining global proposals or longer-term perspectives.

We underline that projections of social security expenditure over the coming decades (up to 2070), in a context of major macroeconomic and demographic changes, may substantially differ from results derived from a purely static framework such as EUROMOD. This limitation also applies, even in the short term, when analyzing non-marginal policy alternatives that may generate significant second-order effects, notably on labor market outcomes. However, the present mandate required the use of a simple analytical toolbox, combining transparency with ease of interpretation, in order to support a preliminary assessment. Although modelling instruments accounting for behavioral responses, partial equilibrium or broader macroeconomic interactions do exist in Luxembourg, they are not fully accessible to social partners and provide limited, if any, insight into longer-term higher-order effects.⁶ Moreover, these models rely on additional assumptions that are themselves subject to debate and are often difficult to communicate in the public arena, which constitutes the primary target of the present analysis. The need for immediately accessible results, together with resource constraints, therefore justifies the deliberately restricted analytical scope adopted here, which was considered sufficient for the social partner’s objectives. Within this context, we relied on this “day-after” tool, which indeed proved adequate to support the formulation of concrete reform proposals during recent discussions with the government on pension amendments implemented from 2026 onwards.⁷ What should remain relevant during necessary additional steps to come.

This study is initially based on a previous one carried out by DIW Berlin (*Ochmann et al., 2014*), yet deviating a bit from it. For example, we simulate cross-border households explicitly. In the DIW report, cross-border households are indirectly considered through proportional summary keys indicating to what extent they may contribute to the “totals”, in case of policy changes, as a proportion of the resident population. Moreover, we build up for the two populations an explicit background information set which may help in understanding outcomes resulting from alternatives (gross income distributions for residents and cross-border households, socio-economic status, share of households through taxable income brackets, etc.).

Finally, although the analysis is specific to Luxembourg, the policy alternatives and methodology considered here may also be relevant for other countries with comparable socio-economic fundamentals, particularly in the context of the EU-wide concern regarding the fiscal implications of population ageing.

The paper is structured as follows. Firstly, we briefly describe the tools (models and data) on which this study is grounded and provide some information on the benchmark “STD” currently in force in Luxembourg, which will serve as a basis for comparisons (Section 2). Then we present the alternatives to be examined (Section 3) and develop the outcomes resulting from these alternatives, addressing the issue of the complexity of the comparisons and attempting to progressively reduce it (Section 4). Finally, we conclude (Section 5).

6. For longer term projections and open-source instruments, see the MiDAS_LU dynamic microsimulation model for Luxembourg in *Dekkers et al. (2022)* and *Liégeois (2025b)*; however, this is targeting the resident population only.

7. <https://cnap.public.lu/fr/actualites/2026/change26.html>

2. The Tools and our Benchmark “STD”

We introduce some basic building bricks that could be useful for a better understanding of the general outcomes resulting from the alternatives that will be presented in the next section. These elements concern the micro-data and the models specifically developed for this study (Section 2.1.) as well as some information on the benchmark that will be used as a reference when analyzing the effects of these alternatives (Section 2.2.). More details and contextual comments on the models can be found in *Liégeois (2025a)*, some extracts of which being reproduced here *in extensor* for the sake of self-sufficiency of the present paper.

2.1. The Models and Data

As developed in *Liégeois (2025a)*: Section 2, would the Luxembourg resident population be alone at stake, EUROMOD (*Sutherland and Figari, 2013*) might be run on the classical “European Union Statistics on Income and Living Conditions” (EU-SILC) data.⁸ Unfortunately, EU-SILC data are not covering cross-border households.

Alternatively, a EUROMOD model has been running for years on the “Household Finance and Consumption Survey” (“HFCS”) data for residents.⁹ And fortunately, HFCS data are also available in Luxembourg for cross-border households, albeit via a separate survey organized by the *Banque Centrale du Luxembourg in collaboration with LISER*, so a model covering this population appearing possible as well.¹⁰ Therefore, we choose to work on those HFCS data in the present study, both for residents and cross-border households, also considering that the HFCS survey for cross-border households “is specifically designed to complement the Luxembourg Household Finance and Consumption Survey” for residents (*Chen et al., 2021*).

Three waves of HFCS data have been collected up to now for Luxembourg: 2013 (income reference year), 2017 and 2020 (the latter being said “coming soon”, at the date of the present study). This is the reason why this paper is based on the most recent wave available when the microsimulation tools were designed (2022), that is 2017 (income reference year).

However, HFCS data show some drawbacks. For example, (monetary) variables are sometimes aggregated or missing, compared to EU-SILC. Moreover, HFCS data are specifically targeting “active” cross-border households (that is, residing in the Greater Region but involving at least one member working in Luxembourg), hence not embedding all persons covered through the LU social security or fiscal systems.¹¹ This may reduce the total public receipts due to Luxembourg, compared to what can be identified through other comprehensive statistical sources, whatever in terms of social contributions or personal income taxes ignored, hence weakening the outcomes in terms of *distributional effects* on that partial population. However, we show in *Liégeois (2025a)*: Section 2 and in Section 2.2. below how to roughly complete the picture to add *social contributions* and *personal income taxes* from those households ignored through the HFCS data; what we call a “macro adjustment”.

Moreover, we have to impute some missing information (for example on incomes, including pensions) in HFCS data for cross-border households, based on some broad approximations.¹²

We are hereafter referring to those data as “HFCS-XB”, for active cross-border households, to be distinguished from “HFCS-R” data for residents. And we use “EUROMOD/HFCS-R” and “EUROMOD/HFCS-XB” to refer either to the EUROMOD/HFCS models based on HFCS data, or to the EUROMOD input databases built from the same datasets, depending on the context.

In the same vein and for simplicity reasons in the present paper, we are referring to “XBs” (cross-borders) as to “active cross-border households”, “all cross-border households” or “cross-border commuters”, indifferently and depending on the context. Moreover, the “total receipts” often evoked

8. <https://ec.europa.eu/eurostat/web/microdata/european-union-statistics-on-income-and-living-conditions>

9. *Kuypers et al. (2016)*; *Kuypers et al. (2020)*, *Chen et al. (2020)* and *Surname et al. (2020)*.

10. <https://www.bcl.lu/en/index.html> and *Chen et al. (2021)*.

11. The “Greater Region” lies at the crossroads of the rivers Rhine, Saar, Meuse and Moselle. It covers 65.401 km² with more than 11.6 million inhabitants from the Grand Duchy of Luxembourg as well as the territories surrounding Luxembourg (Lorraine in France, Wallonia in Belgium, Saarland and Rhineland-Palatinate in Germany) where most cross-border workers come from (<https://www.granderegion.net/en/The-Greater-Region-at-a-Glance>).

12. See ; *Liégeois, 2025b* : Section 2.

are here targeting specifically the total public (social contributions and personal income taxes) financial receipts from personal labor and non-labor earnings.

Therefore, for the sake of the present study and given our objective to involve the XBs in the analysis as well, we have to update a former version of the EUROMOD/HFCS-R model, to take into account a more recent wave for the HFCS data for residents, namely the 2017 one (income reference year), and more recent policy systems in the EUROMOD model as such, up to 2017. And we build a new EUROMOD/HFCS-XB model targeting that additional population.¹³ Those EUROMOD/HFCS models assume full benefit take-up and no tax evasion (households are considered, not firms). By relying on HFCS data, the resulting estimates are also expected to be more accurate for higher income deciles and for wealth-related variables than those derived from SILC-based models.

Finally, most outcomes in this study resulting from the “*EUROMOD/HFCS models for Luxembourg resident and cross-border populations, version 14.62+ Beta release (3.4.10), in combination with the author’s computations*”, we are considering this piece of information as implicit for all tables where no other mention of sources is explicitly provided.

2.2. An Overview of Outcomes for the Benchmark “STD” (2017)

Table 1 gives an overall view of the benchmark “STD” which is a picture available for Luxembourg in 2017 (income year), as derived from EUROMOD/HFCS-R and EUROMOD/HFCS-XB, and completed with regard the macro adjustment which is dealing with “non-active” XB households who are not covered by HFCS/XB.¹⁴

Let’s just mention here that social security contributions and personal income taxes paid by XBs in Luxembourg are not proportional to the respective populations.¹⁵ The global tax rate on XBs’ income (10.4%), is also much lower than for residents (17.2%). This is explained by background fundamentals which differ between the two populations.

For example and with regard employees, the most frequent “employment status” both for residents (43%) and for active cross-border households (56%), **Figure 1** is showing up the distribution of wages (EUROMOD variable “yem”).¹⁶ Those are classified based on the so-called “Minimum Social Wage (MSW) / *Salaire Social Minimum*” which is an important social parameter in Luxembourg. For example, the monthly base for computation of social contributions is most often limited to 5*MSW: above this upper limit, no supplementary contributions are requested. The MSW represents 1,998.59 EUR per month in 2017. Clearly, XB employees are more often facing lower wage levels than residents.

More largely, **Figure 2** is classifying the fiscal households based on fiscal brackets and the main “classes of tax”. The latter can be identified as “1” (roughly “single without dependents and aged < 65”) and “2” (roughly “couples”) and the less frequent “1a”-class (roughly “single with dependents or aged ≥ 65”). Fiscal households are all persons belonging to the same dwelling and considered as to be taxed jointly on income, taking into account the relevant fiscal rules. “Class-2” households

13. Other approaches dealing with cross-borders are referred to in **Liégeois (2025a)** : Section 1). For example, **Sologon et al. (2023)**, are “considering the situation of cross-border workers in a cross-national comparison of incomes”, including Luxembourg, and “develop a spatial microsimulation model of the cross-border region”. Their approach “relies on combining census data, with EU-SILC [administrative and HFCS data] and with an income generation model which incorporates the complexity of the tax-benefit rules of four systems (Luxembourg, Germany, Belgium and France) via the EUROMOD microsimulation model”. This promising development, which involves statistical matching from several sources, could show some synergy with the present study.

14. Basically, **Liégeois (2025a)** - Section 4.6 considers that XB households not covered by the HFCS-XB survey (hence the EUROMOD/HFCS-XB model) are composed of pensioners and their non-active relatives and hypothesizes that pensions received by those pensioners broadly correspond to those paid by Luxembourg to all present and past XBs. Finally, social contributions and personal income taxes are computed on that basis, with for the latter an average tax rate (after tax credits) identical to the one active XBs are facing. Under the assumption that the non-covered population consists of pensioner households, and provided that this prior is well grounded, the contribution of this group to total public revenues in Luxembourg remains limited (around 1% of the total). Pensioners are exempt from several types of social contributions, in particular pension contributions. Consequently, potential income measurement errors for this group are not expected to affect the overall results that much.

15. We are interested in total receipts for Luxembourg only. Therefore, outcomes from EUROMOD/HFCS-XB are primarily based on total incomes, whatever originated from Luxembourg or other countries, given that the tax rate is fixed on that basis by principle, before being applied on a cross-border fiscal household level to the sole income coming from Luxembourg (EUROMOD variable “yem_lu”) to derive the income taxes due to that country specifically. See **Liégeois (2025a)** Section 2.

16. **Liégeois (2025a)**: Graph 1.

represent 37% of fiscal households in the resident population (54% for active cross-border households), "Class 1" 47% of households for the residents (38% for active cross-border households). Here again, we can see that XB active households are more concentrated around the lowest tax brackets.

The next section in **Table 1** is reporting about several "well-being" and inequality indices, all dimensions genuinely embedded in an approach undertaken through microsimulation.

We consider here as an indicator of individual "well-being" the so-called and standard equivalent income which is the ratio between the disposable income of the household (= gross income + social benefits – social contributions – personal income taxes) and a coefficient considering the composition hence needs of that household.¹⁷ The equivalent income is determined at the residence household level -all persons belonging to the same dwelling- and then each member of the household is attributed this household value.

Then, the at-risk-of-poverty rate (hereafter the "poverty rate" or "poverty") represented here is the share of population below 60% of the median equivalent income, a threshold called the "poverty line".

Another indicator is telling something about the unequal distribution of well-being throughout the population: the inequality Gini coefficient (hereafter the "Gini"). As summarized in **Vergnat et al. (2022)**, this coefficient is an index with a value between 0 and 1 (0.2993 in Luxembourg in 2017, based on EUROMOD/HFCS-R), increasing if inequalities in equivalent incomes become greater (zero would indicate perfect equality; the same income for all). It is equal to the absolute Gini index (the average absolute difference between incomes, 2,103 EUR/month) divided by twice the average equivalent income (3,513 EUR/month).

Table 1 is dealing with both residents and XBs and outcomes could be broken down based on usual basic typologies for sub-groups of population: for example the gender, the type of household, the decile of equivalent income and the class of age.

We are not commenting the levels of well-being, poverty and Gini resulting from the EUROMOD/HFCS-XB platform for XB active households, shown up in **Table 1** as well. This population is partial only and less information is collected, compared to the usual reference involving a whole (resident) population, hence comparisons with residents being more hazardous, as mentioned in the introductory section.¹⁸ At most might those results be of interest when assessing the effect of an alternative *within* the cross-border population itself, i.e. while comparing alternatives and not populations, what is done in the full reports, including **Liégeois (2023a)**.

Those above-mentioned indicators will serve as references while analyzing the effects of alternative socio-fiscal policies, compared to our benchmark "STD", in **Section 4**.

Table 1. An overall view of the BENCHMARK "STD" (income year 2017) for resident and XB households

Data and EUROMOD platforms	SILC *	HFCS-R	HFCS-XB (Active households & LU-incomes only)
Population covered by the survey, in persons	574,184	535,897	418,997
Taxable Income, before Tax allowances, in millions € / year (from tinty_s in HFCS-R, or tinty_lu_s for Active XB households)	20,808	20,041	7,947
... out of additional amount from XBs not covered by the survey (for subsequent macro adjustment, see Section 2.4)			1,109
Public Revenue (in millions € / year)			

17. This so-called OECD-modified equivalence scale is attributing a weight of 1 to the first adult, 0.5 to additional persons (aged fourteen or more), and 0.3 per child (aged under fourteen).

18. See **Liégeois (2025a)** Section 2.2 and Appendix A : "Less information is collected, compared to HFCS-R. Some variables are missing or have been aggregated [and] we have to impute some missing information in HFCS-XB".

Data and EUROMOD platforms		SILC *	HFCS-R	HFCS-XB (Active households & LU-incomes only)
Social contributions		3,914	3,714	2,030
	Employee	1,820	1,693	945
	Self-employed	178	221	27
	Others (Long term care from Social assistance)	3	3	0
	Employers	1,710	1,667	1,059
whose	Credited (Replacement income, Social assistance, Pensions)	202	130	0
... out of additional amount from XBs not covered by the survey (for macro adjustment)				62
Personal Income Tax		3,289	3,439	824
	Implicit Tax Rate on Tax Base (tiny_s – tinta_s, or LU versions if XBs), before Tax Credits		20.9%	14.4%
	Global Tax Rate (after Tax Allowances & Credits) = Income Tax / Taxable Income, on average		17.2%	10.4%
... out of additional amount from XBs not covered by the survey (for macro adjustment)				115
► Total Public Revenue (before macro adjustment)		7,203	7,153	2,855
Inequalities		If All Incomes		
Gini				
	Relative = Abs / (2*Avg)	0.2524	0.2993	0.1940
	Absolute (in € / month)	1,701	2,103	1,088
	Average (in € / month)	3,371	3,513	2,804
Poverty				
	Line (in € / month)	1,790	1,770	1,564
	Rate	11%	13%	2%
	by Type of Residence Household:			
	Single (<65)	14%	16%	1%
	Single (65+)	8%	9%	2%
	Single with dependent(s)	31%	18%	0%
	Couple - 0 dep	6%	5%	2%
	Couple - 1-2 dep	11%	18%	2%
	Couple - 3+ dep	14%	21%	2%
"Well-being" , as equivalized income (all in € / month), on average				If All Incomes

Data and EUROMOD platforms	SILC *	HFCS-R	HFCS-XB (Active households & LU-incomes only)
All	3,371	3,513	2,804
1st Decile	1,574	1,472	1,635
by Type of Residence Household:			
Single (<65)	3,308	3,249	2,828
Single (65+)	3,215	3,440	2,878
Single with dependent(s)	2,473	2,661	2,530
Couple - 0 dep	3,922	4,150	3,194
Couple - 1-2 dep	3,117	3,292	2,768
Couple - 3+ dep	2,673	2,826	2,353

Source: Liégeois (2025a) Table 2

*A comparison with official statistics can be found in the EUROMOD country report for Luxembourg (<https://euromod-web.jrc.ec.europa.eu/resources/country-reports> - Archives - I3.0+ - Annex 2: Validation tables)

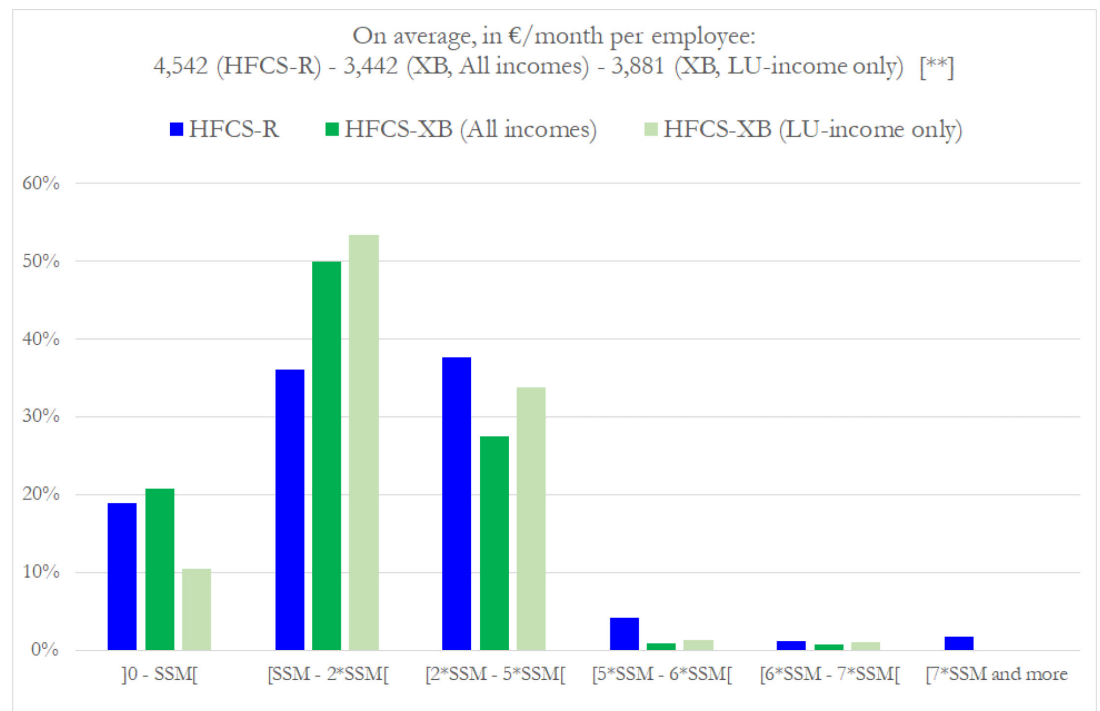


Figure 1 Share of employees, based on their average earnings compared to the MSW, in % of employees with positive (>0) labor earnings [*] (MSW = 1998.59 EUR / month in 2017).

Source: Liégeois (2025a): Figure 2A). [*] The averages are whatever the working time (part-time and full-time workers altogether), which partially explains discrepancies with Clément et al. (2023) outcomes on pages 50 on. [**] The "All incomes" column involves all XB employees with positive employment income (yem > 0 and "employee", as referred to in EUROMOD through the labor economic status variable "les" == 3); "LU-income only" is dealing with that part of XB population earning employment income coming from Luxembourg (yem_lu > 0), the latter group being embedded in the former one, hence a sub-group of it

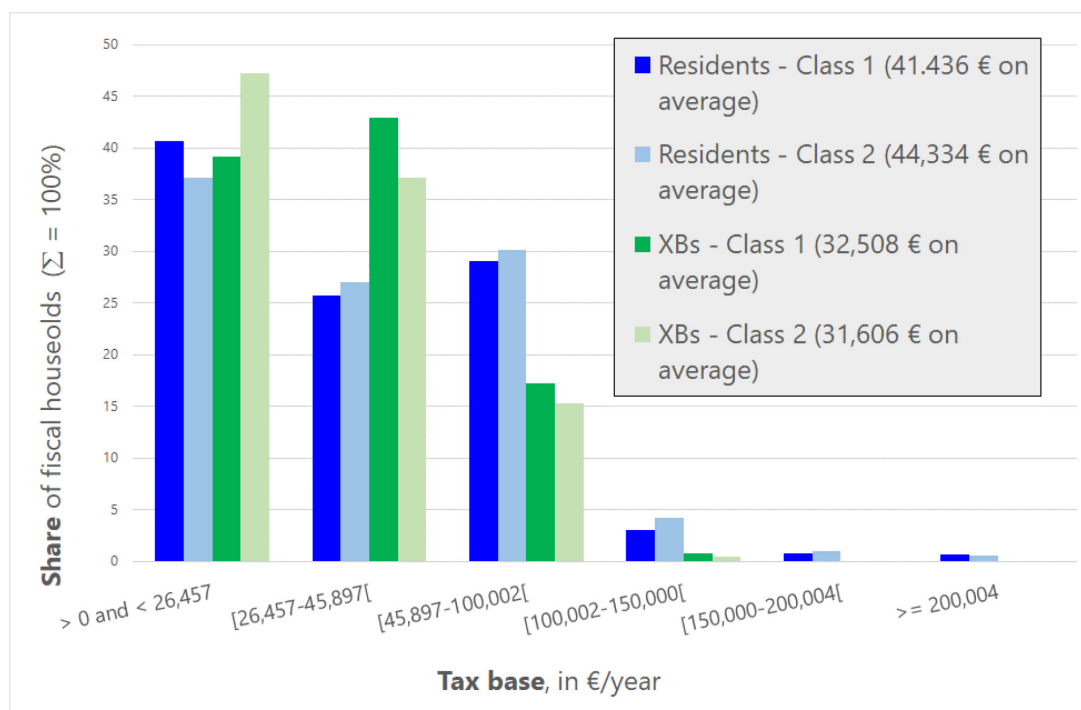


Figure 2 Share of fiscal households across the fiscal brackets, based on their yearly tax base, if >0 and CLASS-1 (single without dependents) or CLASS-2 households (couples) (as resulting from EUROMOD /HFCS-R and EUROMOD /HFCS-XB).

Source: Liégeois (2025a): Graph 4). Note to the reader: it should be understood from the “27%” seen for bracket [26,457-45,897[for resident couples (class-2 fiscal households) that 27% of those households are experiencing a tax base situated between 52,914 €/year (twice 26,457) and 91,794 €/year, given that this tax base is divided by 2 in Luxembourg for class-2 households, before applying the tax schedule for computing the tax on income (finally multiplied by 2 for an effective value).

3. Alternative Systems of Socio-Fiscal Policies

Departing from “STD”, the standard system of socio-fiscal policies implemented in Luxembourg for the (income) year 2017, we are now introducing a large number of alternatives that modify some parameters, like the maximum thresholds for the computation of social contributions, the tax schedules, etc.).

Tables 2–4 summarize the alternatives discussed in this paper, considering alterations of the social contributions for pensions, health care (in-kind) or long-term care (dependency), as well as modifications in the income tax schedule. For each of them, we provide some information about its contents. 2-letter acronyms have been chosen for those alternatives. The first letter in the code is underlying the policy considered (Pensions / Health / Dependency / Income tax) and the second letter is the order number of the alternative within each category (1,2, ... , 9, A, B, ... H).

Table 2. Alternative systems of socio-fiscal policies - 1st part STD, dependency and health care

#	Underlying Policy	Ref.	Contents (all in relation with Social Contributions)*
1	BASILINE	STD	EUROMOD simulation of System presently in force "LU_2017_STD"
2		D1	Deductibility (for the computation of social contribution base) from 0.25*MSW in STD down to 0% ‡
3	Dependency (Long Term Care) (target cf. health/in-kind, out of pensioners) (7% of social contributions in 2018) †	D2	Deductibility (for the computation of social contribution base) from 0.25*MSW up to 0.5*MSW
4		D3	Contribution rate increase from 1.4% in STD up to 1.6% of labor income (to roughly balance the higher deductibility above in [D2])
5		D4	Contribution rate increase from 1.4% up to 2.4%
6		D5	[D2] + [D3]: Increase deductibility and balance it through a higher contribution rate
7		H1	Max threshold for the social contribution base from 5 MSW in STD to 7 MSW (see Section 2.2 and ‡)
8		H2	No threshold for the social contribution base (NB: upper bound set to <u>1000 MSW</u> , practically)
9		H3	Reduced <i>fiscal deductibility</i> of social contributions: from 100% (not taxable in STD) down to 50%
10		H4	Social contribution rates from 2.8% in STD to 3.8% (hence +2% if employ-ees/-ers altogether)
11	Health care / in-kind (Social Contributions, including "credited", for employ-ees/-ers, self-employed, replacement income, pensioners, social assistance) (29% of social contributions in 2018) †	H5	Progressivity of social contrib. rates: 2.8% (resp. 3.8%) for that part of labor income < 2 (resp. >=2) MSW
12		H6	Progressivity of social contrib. rates: 1.8% (resp. 3.8%) for that part of labor income < 2 (resp. >=2) * MSW
13		H7	[H5] + [H1]: Progressivity (2.8%/3.8%) and max threshold to 7 MSW
14		H8	[H6] + [H1]: Progressivity (1.8%/3.8%) and max threshold to 7 MSW
15		H9	[H5] + [H2]: Progressiv. (2.8%/3.8%) and no threshold
16		HA	[H5] + [H2] + [H3]: Progressivity (2.8%/3.8%), no threshold and fiscal deductibility to 50% only

*For details about "STD" and reference parameters or policies, refer to Islam et al. (2020).

†Source: EUROMOD Country Report for LU, Y12 (2018-2021), Table 4.8 (page 73), Employee / Self-employed, Employer and Credited contributions, from External sources and for Residents only.

‡Minimum Social Wage (MSW, Section 2.2): 1.998,59 EUR/month as on 1 January 2017. In STD, 25% of this amount is deducted from the social contribution base (gross labor income) before deriving social contributions (some % of the base).

Table 3. Alternative systems of socio-fiscal policies - 2nd part - pensions

#	Underlying Policy	Ref.	Contents (all in relation with Social Contributions)*
17	Pensions (Social Contributions, including “credited”, for employees, self-employed, replacement income) (60% of social contributions in 2018) Source: cf. [***] in Table 2	P1	Max threshold for the social contribution base from 5 MSW in STD to 6 MSW †
18		P2	Max threshold from 5 MSW to 7 MSW
19		P3	No threshold for the social contribution base (NB: upper bound set to 1000 MSW, practically)
20		P4	Social contribution rate from 8% in STD to 9% both for employees and employers (hence +2% altogether)
21		P5	[P4] + [P2]: Rate from 8% to 9% and max threshold to 7 MSW
22		P6	[P4] + [P3]: Rate from 8% to 9% and no threshold
23		P7	Progressivity of social contribution rates: 8% (resp. 9%) for that part of labor income < 2 (resp. >=2) * MSW
24		P8	Progressivity: 7% (resp. 9%) for that part of labor income < 2 (resp. >=2) MSW
25		P9	[P8] + [P2]: Progressivity (7% and 9%) and max threshold for the social contribution base to 7 MSW
26		PA	Progressivity: 9% (resp. 10%) for that part of labor income < 2 (resp. >=2) * MSW
27		PB	[PA] + [P3]: Progressivity : 9% (resp. 10%) and no threshold
28		PC	Social contribution rate from 8% to 12% for employees/-ers
29		PD	CSV: cf. [P1] + Contribution Solidarité Vieillesse / CSV: 5% flat rate with the Minimum Pension (“MP” = 1771.75 €/month in 2017) deducted from the contribution base (pension income) at individual level ; no deductibility from income tax base & No Threshold for CSV
30		PE	CSV: cf. [PD], but 25% of MP deducted from the contribution base (pension income) only
31		PF	CSV: [PD] + [P6] (social contribution rate to 9% and no max threshold)
32		PG	CSV: cf. [PD], hence [P1], but Proportional Tax for CSV (flat, but increasing with class of pension income): 0% if social contrib. base <= MP, 1.4% if base in]MP, 2*MP], 2.8% if base in]2*MP, 3*MP] and 4.2% if base > 3*MP
33		PH	CSV: cf. [PD], hence [P1], but Progressive Tax for CSV (cf. Income tax), same rates and limits as in PG

*For details about parameters or policies in our benchmark “STD”, refer to Islam et al. (2020).

†Minimum Social Wage (MSW, Section 2.2): 1.998,59 EUR/month as on 1 January 2017.

Table 4. Alternative systems of socio-fiscal policies - 3rd part - income taxes

#	Underlying Policy	Ref.	Contents (all in relation with <i>Income Taxes</i>)*
34		11	Brackets unchanged but higher Rates for higher Brackets : 45.897 € and more (40%, rather than 39% in STD), then 42% (40% in STD), 44% (41% in STD) and 46% for higher brackets (42% in STD)
35		12	More higher brackets and higher Rates : 39% up to 100.000 € (100 K), 40% up to 150 K, 41% up to 200 K -up to here as in STD-, then 42% up to 300 K, 44% up to 400 K, 46% up to 500 K, 48% up to 600 K, 50% up to 700 K, 52% up to 800 K, 54% up to 1000 K and 56% over 1000 K
36	Personal Income Taxes (Tax schedule) (equivalent to 86% of social contributions in 2018) Source: cf. ¹⁹ in Table 2	13	Idem [11] but "class-1a" (Section 2.2) dropped and merged to class-2
37		14	Lower Rates for lower (unchanged) brackets : 0%, 2%, 4%, 6%, 8%, 10%, 12%, 14% and 17% respectively for the 9 first tax brackets (0%, 8%, 9%, 10%, 11%, 12%, 14%, 16% and 18% in STD), then unchanged compared to STD
38		15	Enlargement of lower Brackets (Rates unchanged): 0 € ; 16,000 € ; 17,000 € ; 18,000 € ; 19,000 € and [20,000 € – 20,625 €] for the 6 th bracket (0 € ; 11,265 € ; 13,137 € ; 15,009 € ; 16,881 € and [18,753 € - 29,625 €] in STD)
39		16	All combined [11] + [14] + [15]: higher Rates for higher Brackets and enlargement of lower Brackets (with lower Rates)

*For details about our benchmark "STD" and reference parameters or policies, refer to Islam et al. (2020).

For a better immediate understanding, we can point out that a maximum threshold of 5 times the Minimum Social Wage (Section 2.2.) is often set in Luxembourg, for example as a basis for derivation of contributions if the worker is not a civil servant (labor income exceeding this limit is not considered for social security contributions).

Moreover, four strong yet quite hypothetical alternatives "C1-C4" have been imagined, which purpose is to look for a combination of systems possibly leading to high additional total receipts (Table 5). The objective was to reach approximatively an equivalent of a 5%-GDP gain in terms of total additional receipts.

Such a target is sometimes cited by social partners in relation with -about half of- the expected increase in age-related expenditure over the long term (see the introductory section). This might be reached through a 25% increase for the sum of social contributions and personal income taxes.¹⁹

We underline that several of the policy alternatives considered are non-marginal and may generate significant higher-order or feedback effects. Such a complementary exercise, given the high level of

19. Roughly evaluated, the Gross Domestic Product (GDP) being 55.3 billion € in 2017 (STATEC, 2019), the social contributions amounting to 6.13 billion € (OECD, 2023; tax revenue in 2017, "2000 Social security contributions") and the taxes on income, profits and capital gains of individuals to 5.06 billion € (OECD, 2023; tax revenue in 2017, "1100 Taxes on income, profits and capital gains of individuals"), a target of 5% of GDP for additional funding would represent $5\% \times 55.3 \text{ billion } € = 2.77 \text{ billion } € / \text{year}$, that is $2.77 / (6.13 + 5.06) = 25\%$ of total social contributions + personal income taxes. This is our reference target with regard to the combined systems C1 to C4.

changes, is mainly theoretical (at most seen as a long-term perspective), aiming at raising the question of a feasibility for those tracks and entering the black box in terms of implications. Nevertheless, the transparent “day-after” approach adopted here is considered sufficient to meet the initial objectives of the analysis, as outlined in *Section 1*.

Table 5. Alternative systems of socio-fiscal policies - Final part combining instruments for a global target

#	Combined Policy	Ref.	Contents	Remarks
40	Combination of policies to target an additional funding of 2.77 billion € / year (EUROMOD-base)	C1	[P6] + ([H2] + [H4]) + ([D1] + [D4]) + I1	“C” (as first letter) for “Combination” of policies, here based on all maximally funding tracks through Pensions (without CSV), Health, Dependency and Income tax <i>Reminder: [P6] = social contribution rate from 8% to 9% & no threshold / [PB] = progressivity 9% and 10% & no threshold / [PC] = 8% to 12% with threshold (5 MSW)</i>
41		C2	[PB] + ([H2] + [H4]) + ([D1] + [D4]) + I1	
42		C3	[PC] + ([H2] + [H4]) + ([D1] + [D4]) + I1	
43		C4	[C1] + [PG]	

4. Towards a Comparative Assessment of Alternatives

The 43 alternatives developed in *Section 3* may generate a large number of indicators dealing with total social contributions and personal income tax, well-being, inequalities and poverty rates, including by population sub-groups and for both residents and XBs.

Table 1, which is part of a description usually proposed by analysts to policy makers, is giving a flavor of such a complexity for a single policy mix (here the baseline “STD”). *A fortiori*, comparing a great number of alternatives is a high dimension challenge that we are now trying to meet by proposing a progressive path of analysis.

As a starter and to ensure the legitimacy of our approach in the eyes of the social partners, we go on with usual practices and indicators well-established in the sphere of socio-economic analyses. While proceeding, we keep in mind the main objective of some additional revenue in terms of social contributions and personal income tax. But we complement this with an additional focus on distributional impacts, with a look at the Gini inequality index and the poverty rate, two other classical indicators. Moreover, while the first one is here logically involving both the residents and the XBs, the next two are addressing the residents only, for reasons evoked in the introductory section.

This limitation to a triple of indicators for a comparative analysis is done to make clearer a combination of objectives. However, we show in the *Appendix A* that those indicators are correlated -hence may also be seen as “related”- to other relevant ones.

We now propose an overall view of alternatives, based on this reduced set of indicators, either separately for each indicator (*Section 4.1.*) or when combining indicators (*Section 4.2.*).

4.1. A View on all Alternatives through the 3 Selected Indicators

Figures 3–6 are showing up an overall view of the 3 basic indicators: the total receipts from residents an XBs, inequalities and the level of poverty among the residents as they result from all alternatives, including the benchmark STD.

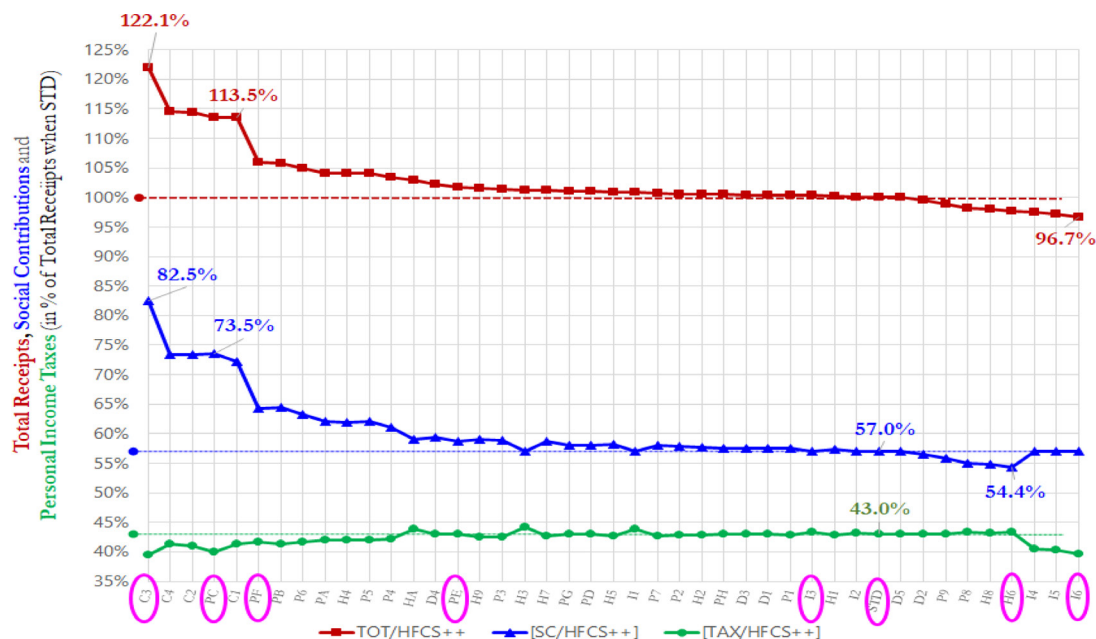


Figure 3 Total receipts for Luxembourg (from residents and all XBs), social contributions and income taxes for all alternative systems of socio-fiscal policies (all outcomes “++” hence including “macro adjustment”, see Table 1) – Ranked based on total receipts

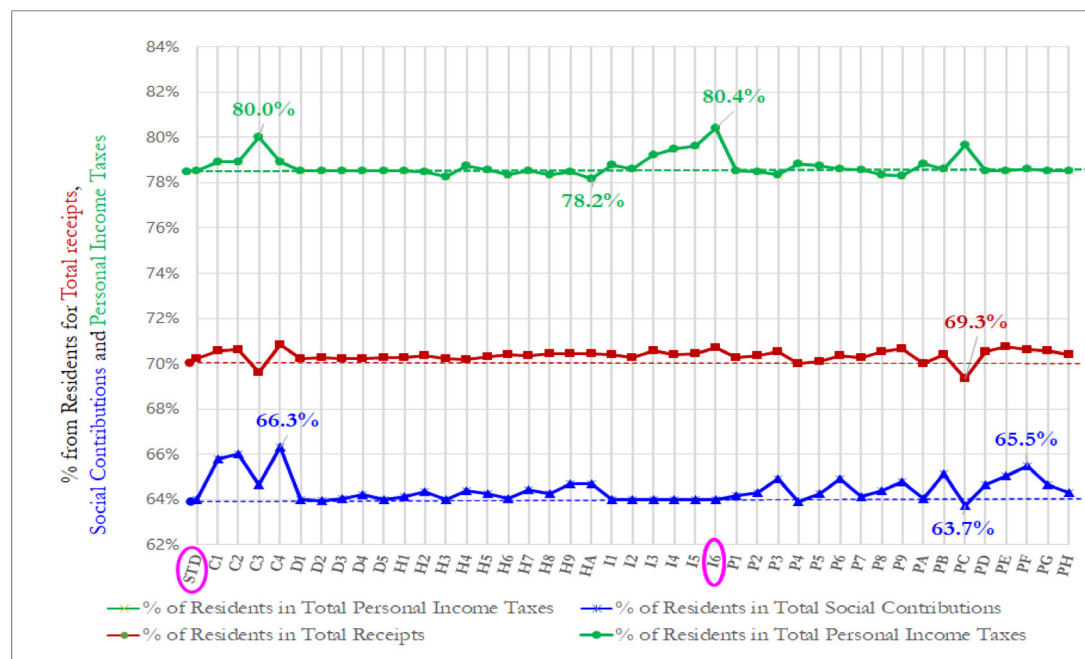


Figure 4 Proportion of receipts for Luxembourg from residents (in % of receipts including “macro adjustments”, see Table 1)

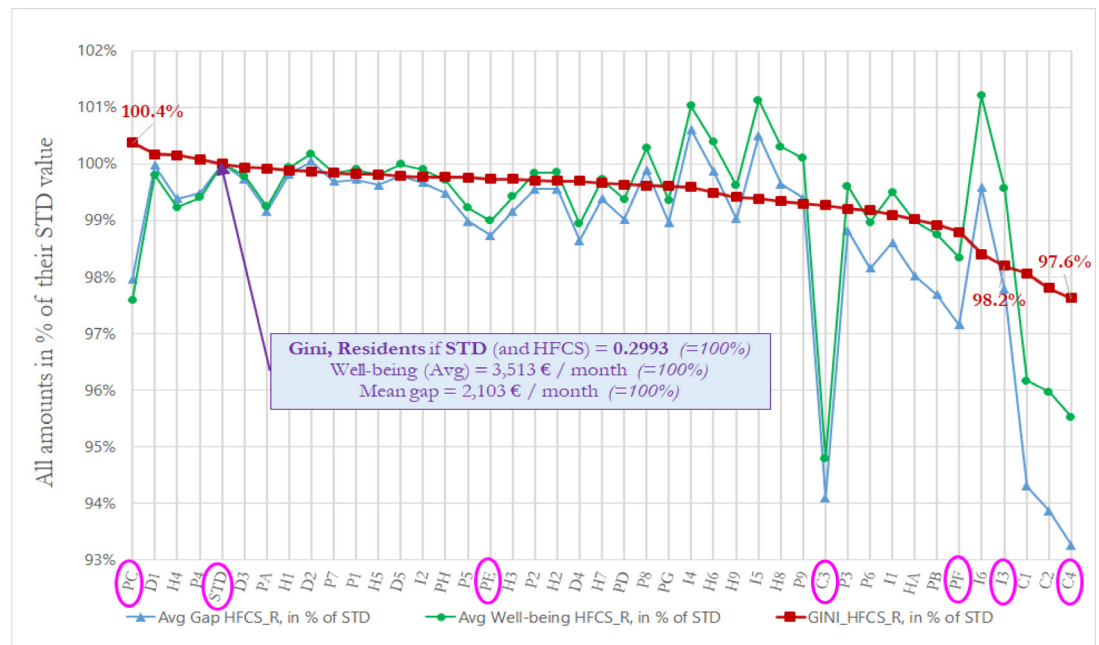


Figure 5 The Gini coefficient for residents and its decomposition for all alternative systems of socio-fiscal policies – in % of their STD value, ranked in decreasing values

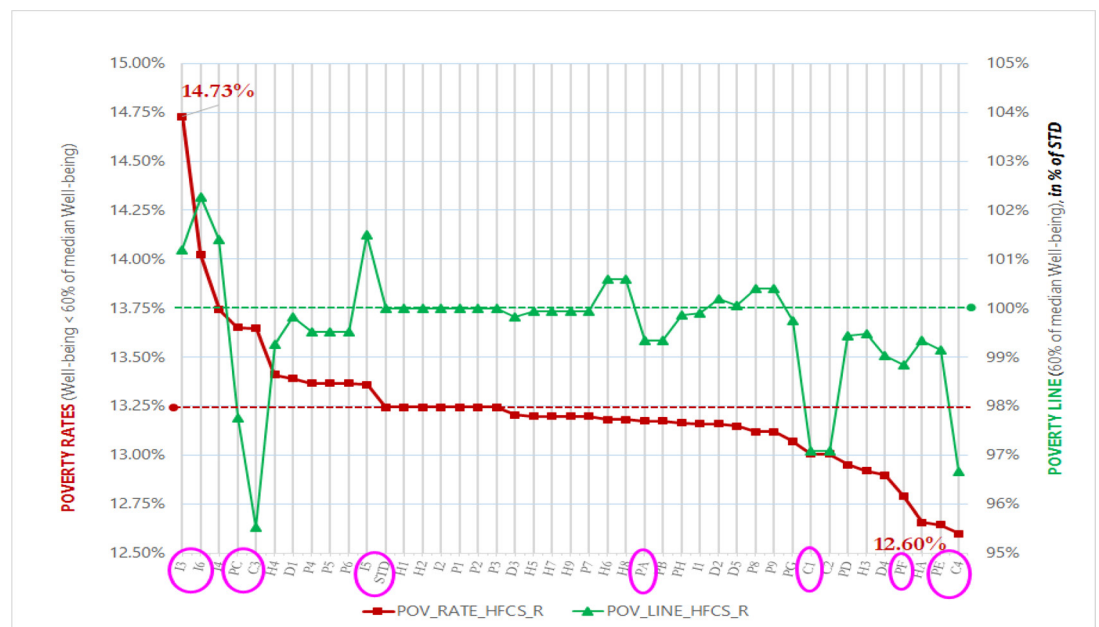


Figure 6 The poverty rate (absolute values) and the poverty line (in % of its STD value) for residents and all alternative systems of socio-fiscal policies – Ranked in decreasing values of poverty rates

Figure 3 shows that among all alternatives, the combination C3 drives to the highest increase in total receipts for Luxembourg (whatever from residents or XBs): 122% compared to its level through the benchmark STD, I6 being the most depressing design with that respect (97%).^{20 21} compared to its level through the benchmark STD, I6 being the most depressing design with that respect (97%). Out of the alternatives "Cx", PC (rate from 8% to 12%) is reaching 114%. A majority of the alternatives are leading to higher receipts, compared to STD, yet just a little for many of them. The three dotted horizontal lines are marking the % relevant for STD.

20. About the ranking of alternatives, see remark in Section 4.2.

21. Such a jump in receipts is not marginal and might induce significant higher order effects. See Introduction and Section 3 for comments.

Gains in receipts are often coming from higher social contributions (blue line), which is directly resulting from the attributes of the alternatives retained by the social partner in the present exercise. Personal income taxes (green line) resulting from alternatives lie usually around or below their STD level, rather, which is partly explained by higher social contributions pushing down tax allowances.

The benchmark STD is reporting 57% of total receipts coming from social contributions, hence 43% from personal income taxes. This can go up to 83% for social contributions in C3 (in proportion of the total receipts for STD) or, leaving aside the combined alternatives, 74% for PC, and down to 54% for H6.

Figure 4 is telling more about the share of receipts coming from residents. In STD, the former are funding around 70% of total receipts going to Luxembourg, 79% of personal income taxes and 64% of social contributions, with some variations throughout the alternatives. For example, I6 is leading residents up to 80.4% for the income taxes. Roughly said, the proportions visible for residents -higher than the share of this population (56%, based on **Table 1**), and still higher for income taxes compared to social contributions - may also be a combined effect of lower earnings for XBs (see **Figures 2 and 3**) and the progressive nature of the income taxation (: **Section 4**).

We now turn to a view of inequalities. **Figure 5** is showing the Gini coefficient pertaining to residents for all alternatives, ranked from the highest value to the lowest one, as well as its decomposition into the average well-being and the average absolute earnings gap (the Gini coefficient is equal to the latter divided by twice the former). All percentages reported are relative to the values of each determinant in the benchmark STD. Very few alternatives are doing worse than STD for the Gini coefficient (>100%), with PC leading to 100.4%. The best option with that respect is C4, which drives the Gini down to 97.6% of what is observed in STD. Everything being constant by elsewhere, those 2.4% would represent a decrease of about 50 €/month for the average (absolute) earnings gap, to be compared to its value of 2,103 € in STD -a gain that might be considered as rather limited by the social planner, hence maybe not a priority-.

The real story in the background differs indeed a bit: we approximatively face through C4 a 7% reduction in the mean earnings gap (blue curve), which is however partially compensated in terms of inequalities by a 5 % loss in the well-being (green curve), altogether leading to this 2% change in Gini. Out of the combined systems, I3 seems the best option, with a reduction of 1.8% for the Gini.

Figure 6 is now giving a flavor of what happens with the poverty rate among residents. This indicator may go up to 14.7% for I3 (13.2% in STD) and can be reduced a little at 12.6% through C4. Like for the Gini, explaining changes in the poverty rate is often a complex task, this being the result of a combination of (changes in) both the median well-being (hence the poverty line which is here 60% of the median) and the distribution of well-being around this line as modified through alternatives. **Figure 6** shows that several configurations can happen: a negative change in both the poverty line and the poverty rate, which seems an intuitive association, like for PA or C1. Or vice versa, as for I5 or I6. But a counter-intuitive increase in poverty combined with a lower poverty line can also be observed, like in PC or C3, which is an indication that the depressing effect of the alternative on the earnings is still more pronounced on lower income deciles than around 60% of the median. In particular, C3 is characterized by a lower Gini index (**Figure 5**) combined with a higher poverty rate. As complementarily shown in **Liégeois (2023a)**, the upper equivalent income deciles are more adversely affected by this alternative than the lower deciles, leading to a compression of the income distribution and, overall, to a reduction in inequality.

Altogether, **Table 6** is summarizing the maximum and minimum (" I_{MIN} " and " I_{MAX} ") reached for the three representative indicators retained as well as the alternative leading to such values, together with " I_{STD} " prevailing for the benchmark in 2017 (STD). The "normalized index" referred to in **Table 6** is clarified in **Section 4.2**.

Table 6. Selected basic indicators and normalized values for STD

Basic Indicator considered β	Min		Max		STD	
	C1-C4 β	I_{MIN}^*	Alternative	I_{MAX}^{**}	Alternative	Normalized index i_{STD} (if Cx included) *
Gini coefficient(residents)	Included	0.2922	C4			
	Excluded	0.2939	I3	0.3004	PC	0.2993
						0.138 *

Basic Indicator considered β	Min		Max		STD	
	C1-C4 β	I_{MIN}^*	Alternative	I_{MAX}^{**}	Alternative	Normalized index i_{STD} (if Cx included) *
	Included	12.6%	C4			
Poverty Rate(residents)	Excluded	12.6%	PE	14.7%	I3	13.2%
Total Receipts for Luxembourg(in millions €/year)	Included	9,849	I6	12,432	C3	10,185
	Excluded			11,558	PC	0.130

*See Section 4.2 - The Gini normalized value g_{STD} directly computed from this table = $(0.3004 - 0.2993) / (0.3004 - 0.2922) = 0.134$ is slightly deviating from the "effective" value 0.138 mentioned in the table, due to rounding effects

4.2. A Look at Alternatives While Combining Indicators

More global and useful lessons can be derived from a combination of these figures. **Figures 7 and 8** are illustrating those associations for the couples "Gini residents – Total receipts" and "Poverty Residents – Gini Residents". We show here a selection of alternatives only, to keep the pictures readable.

Yet PC being the best alternative for *total receipts*, out of the combined "Cx" systems and as visible from **Figure 3**, **Figure 8** is revealing that this is also at the expense of both an increase in poverty and a higher Gini, compared to STD. If we concentrate initially on *poverty* rather, PE (the best alternative with that respect, out of "Cx" ones) is also decreasing the Gini (**Figures 5 and 8**) but with a moderate effect of total receipts only, as also shown through **Figures 3 and 7**. Taking now into account the third dimension, alternative I3 improves the *Gini* at best, yet implying a strong increase in poverty (**Figure 8**) and all this for unchanged total receipts (**Figure 3**).

Therefore, the natural question that now arises is how to find our way through so many alternatives and indicators, in search of a path that would "best" combine the 3 selected dimensions. Let's emphasize already alternative PF that seems of interest for all three dimensions (**Figures 7 and 8**).

Note that summarizing, and ranking up to a certain extent, alternative systems of socio-fiscal policies is neither an easy task nor an ultimate answer. Obviously, an alternative increasing a tax or contribution by 2% will most often drive -unless strong non-linearities due for example to a threshold effect- to results qualitatively close to, yet more pronounced than, another change of the same rate by 1% only. For example, the alternative PC (social contribution rate from 8% to 12% for pensions) will impact indicators in the same way, qualitatively, as P4 (8% to 9%), yet with more pronounced effects, what can be seen for example from **Figure 8**. This does not imply as such that, when outcomes are considered desirable, the former track is better than the latter that might be anyway preferred for any political economy reason: progressive reform, acceptable thresholds or rates, political agenda, ...

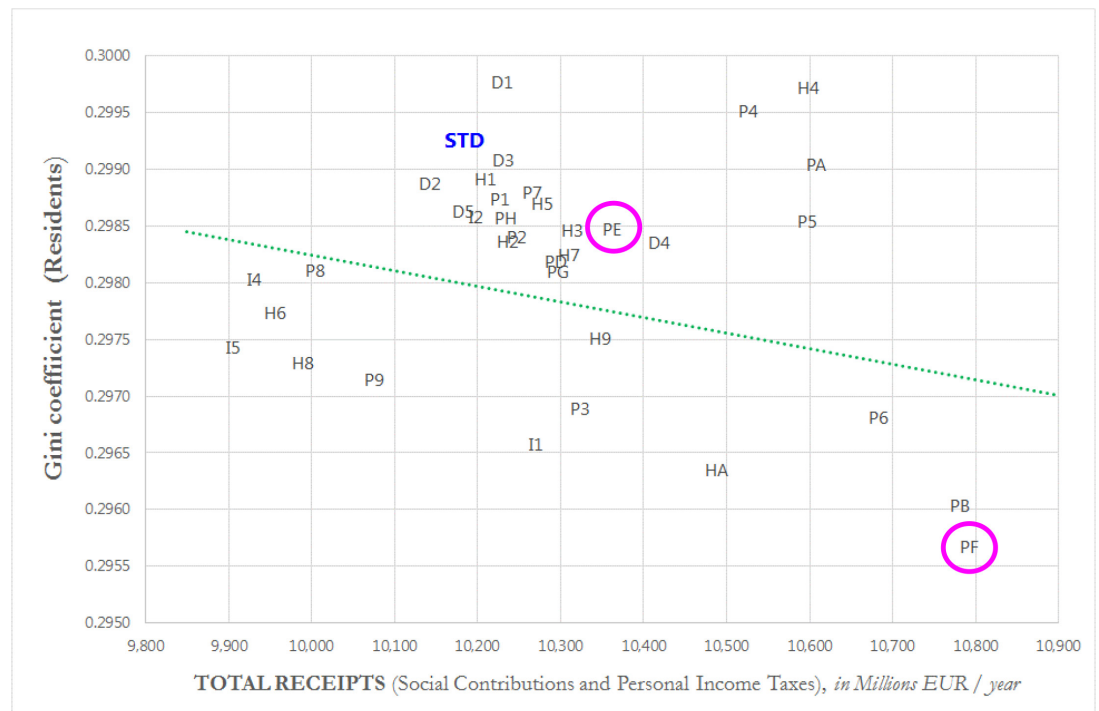


Figure 7 Gini coefficient for residents versus Total receipts, across a selection of alternative systems of socio-fiscal policies

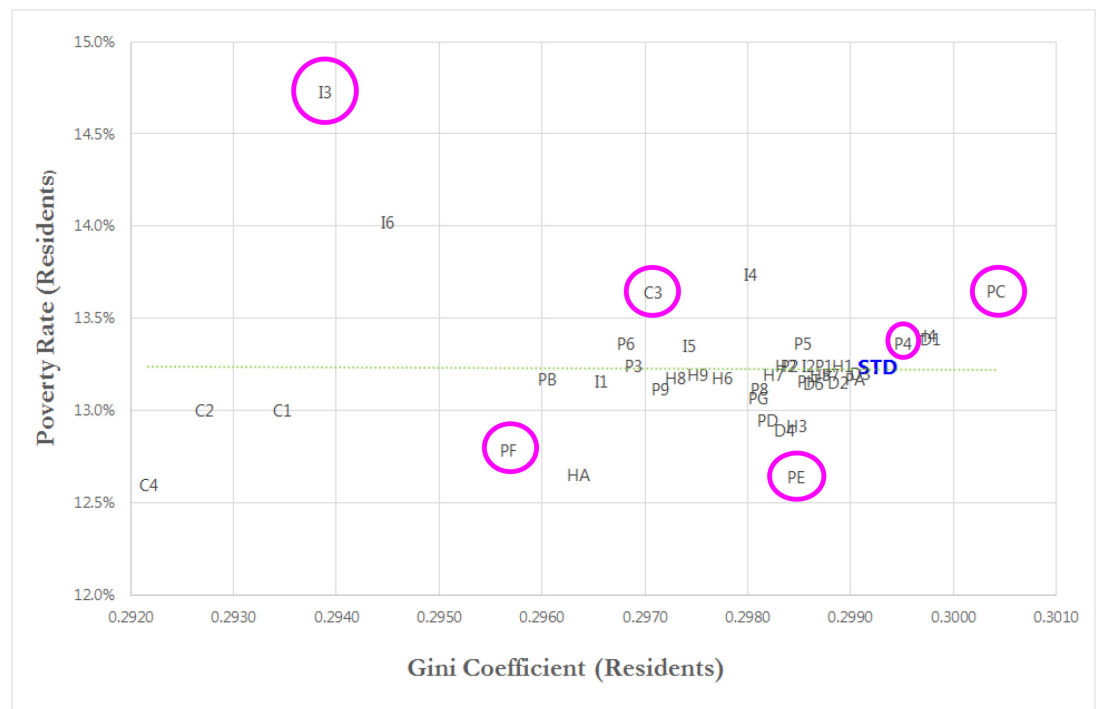


Figure 8 Poverty rate for residents versus Gini coefficient for residents, across a selection of alternative systems of socio-fiscal policies

Nevertheless, we may desire to complete our detailed overview with some assessment about the overall performance of each alternative and a comparison of all alternatives based on such an overall view designated here as the “global performance index” (hereafter, “GPI”).

A simple GPI might be a composite indicator derived from the specific dimensions previously selected: inequalities among the residents (represented by “G”), poverty rates for the residents (“P”) and total receipts (“R”). A vast literature exists on composite indicators, but it lies outside the scope of

the present request from the social partner, as it is not directly suited to exchanges with third parties.²² Nevertheless, the index proposed here may help to clarify internally specific policy advantages or to confirm the relative positioning of alternatives already identified in previous analyses. For these reasons, this type of analytical approach is only briefly mentioned in the present paper.

Practically and taking into account that ranges of values for the basic indicators G, P and R can differ a bit in absolute terms, we may choose to normalize each of them to the interval [0-1]:

$$i_A = \frac{I_A - I_{MIN}}{I_{MAX} - I_{MIN}} \text{ if } R, \text{ or } \frac{I_{MAX} - I_A}{I_{MAX} - I_{MIN}} \text{ if } G, P$$

where:

- A is the alternative system examined (for example STD or P7)
- I is the original value for the basic indicator under consideration here (G, P or R - see remark in the next paragraph)
- i_A is the normalized value of the basic indicator considered for alternative A
- MIN and MAX are the minimum and maximum values reachable for each basic indicator G, P or R throughout all the alternatives (for example, 12.6% to 14.7% for the poverty of residents, with an STD value equal to 13.2% - see **Figure 6**)

The normalized indicator i_A is expected to *increase* when the outcome is socially considered as “desirable”, hence a specification differing for R, on the one side, G and P on the other side (the normalized indicator for P should be greater if P is lower). The higher the normalized index, the better is the original index, socially speaking: “1” is the maximum/best (resp. “0” the minimum/worst) accessible value among all alternatives. Therefore, a value closer to 0 for i_{STD} , like total receipts or the Gini coefficient in **Table 6**, is showing that other alternatives can offer some room for “improvement”, from STD to the best option, compared to the worst one.

We are now properly equipped for combining the basic indicators. At this stage, we can use linear and non-linear approaches. Linear aggregation of indicators assumes full compensability between dimensions, allowing poor performance in one dimension to be offset by good performance in another. When such substitutability is undesirable, non-linear aggregation methods—such as geometric means or non-compensatory indices—provide a more appropriate framework by penalizing unbalanced performance and limiting trade-offs between dimensions.

For sake of illustration here, we consider a linear weighting procedure: $a \cdot g + \beta \cdot p + g \cdot r$, where a , β and g are exogenous parameters revealing the social planner’s preference. For example, $0.1 \cdot g + 0.2 \cdot p + 0.7 \cdot g$ would imply that a stress is put on receipts in the global performance of an alternative. Note that a , β and g are summing up to 1 without loss of generality. Such a linear specification is simple enough and probably sufficient as a first step for a better understanding of the overall procedure.

Appendix B is showing up the first- and second-best alternative systems of policies, as well as the worst one, identified through their global performance index GPI, taking into account several values for the weights a , β and g . For example, a balanced objective with weights 0.3, 0.3 and 0.4 respectively, is leading to PF as the best possible alternative... as evoked earlier.

5. Conclusions

This paper presents alternative, sometimes merely exploratory, avenues devised by a key social partner for the financing of social security in Luxembourg and their implications in terms of income distribution. We build on an earlier analysis carried out by DIW in 2013-2014 but rewrite its content in order to i) add an in-depth examination of the cross-border population in addition to the resident one, ii) test new combinations of socio-fiscal policy systems and iii) isolate determinants important to the results.

The cross-border population plays a more important role in Luxembourg (and on public finances) than in most other developed countries, and therefore definitely deserves to be included for a more comprehensive analysis. The new light shed on Luxembourg by this paper requires the updating of an older version of a EUROMOD model using HFCS micro-data for the resident population, and the

22. See *Nardo et al. (2008)*, *Mazziotta and Pareto (2011)*; *Mazziotta and Pareto (2016)*, *Greco et al. (2019)*.

creation of a new model for XBs. The constraints inherent in micro-data prompts us to opt for 2017 (income year) as the analysis year.

Given the significant structural differences between residents and XBs in terms of socio-economic status, gross labor income and taxable income, we show that total receipts from residents are greater than those from XBs, even when controlling for population size. For the same reason, a reform of the socio-fiscal system in Luxembourg might not have the same impact on XBs (facing lower income levels on average) and residents.

By examining the 42 alternatives devised here (in addition to the standard STD, in force in Luxembourg in 2017), some of them are highlighted, based on considerations including the values achieved for a triple of indicators retained for this overall analysis: total receipts (including XBs), the Gini inequality coefficient and the poverty rate (the latter two for the resident population only).

Table 7 summarizes this point and highlights the scope for change in relation to the reference STD, while considering all the alternatives discussed in the study. The final choice of a specific alternative would obviously depend on the desired level for each indicator, the trade-off between possibly competing dimensions and, more generally, on social preferences or political feasibility.

Table 7. A few alternatives emphasized throughout this study

Criteria and Main values	If considering the combined alternatives C1-C4 (see also Table B1)	If excluding C1-C4 (see also Table B2)
Total Receipts (social contributions + personal income taxes ; residents + XBs) [Min: 8,849 - STD : 10,185 - Max : 12,432], in million € / year	C3: best C4: 2 nd best I6: the worst alternative (but that can also be shown as the best for well-being)	PC: best PF: 2 nd best
Poverty rate among Residents (proportion of persons under 60% of the median equivalent income) [Min: 12.6% - STD: 13.2% - Max: 14.7%]	C4: best PE: 2 nd best	PE: best HA: 2 nd best
Gini inequality coefficient among residents (mean absolute equivalent income gap divided by twice the average equivalent income) [Min: 0.2922 - STD : 0.2993 - Max : 0.3004]	C4: best C2: 2 nd best	I3: best I6: 2 nd best
General Performance Index (GPI)	C4: best <i>balanced</i> (if weights = 0.3/ Gini, 0.3/poverty, 0.4/receipts) C2: 2 nd best <i>balanced</i>	PF: best <i>balanced</i> (if weights = 0.3/ Gini, 0.3/poverty, 0.4/receipts) HA: 2 nd best <i>balanced</i>

While reviewing the results highlighted in this paper, the reader should bear in mind important aspects that may also merit further examination in the future.

Firstly, the 42 (+1) alternatives selected for this paper are mainly *ad hoc* (in particular, the changes are parametric) and mainly based on the CSL's concerns and proposals.

When designing the reforms, other choices could have been made for several parameters (thresholds, rates). To a certain extent, the results would be easily resized if other parameter levels were preferred (such as 8.5% for social contribution rates for pensions, instead of the current 8% in STD and 9% in alternative P4). If we consider that the social and tax system is sufficiently linear (hence smooth enough) and additive (which is clearly a simplifying assumption), turning to 0.5% for a change, rather than 1%, could simply lead at first sight to results half as important (at least for social contributions).

Secondly, the method we have chosen to reduce the high dimensionality of the comparative analysis is also partly *ad hoc* and crude. It includes both the selection of core indicators from a subset of classical ones, while considering their correlation matrix (Section 4.1. and Appendix A), and the design of a Global Performance Index (Section 4.2.). But this particular approach is conceived here as simple enough and probably sufficient as a first step towards a better overall understanding of the analytical challenges involved.

Finally, several alternatives would lead to "major" changes in socio-fiscal policies. Some of them, such as PC or C3, were mainly designed to provide an initial sense of the implications, assuming that any future increase in pension spending would be financed solely through additional resources while

keeping in mind, yet not addressing at first sight, downstream effects. However, such a shift raises two types of questions.

The first question focuses precisely on the possible effects of alternatives that would go beyond the immediate or “day-after” impact, including behavioral responses, partial or general equilibrium effects and longer-term perspectives.

In particular, some alternatives may have substantial effects on labor supply if they alter net wages, rather than merely increasing charges borne by firms. Future research could analyze these behavioral responses, once policymakers have identified preferred alternatives on the basis of this preliminary step, through a discrete-choice modelling approach, similar to that used for Luxembourg by **Berger et al. (2011)**. Moreover, higher gross wages may reduce labor demand, with possible effects on both resident and cross-border workers, thereby raising a partial-equilibrium issue. Finally, given the “lumpy” features of Luxembourg’s labor market—participation decisions, cross-border commuting costs, and fixed working hours—a Random Utility Random Opportunity (RURO) framework could also be considered, insofar as data constraints allow.²³

However, the need for sufficient confidence, on the side of the social partners, in the proposed modeling apparatus leads us to a static platform such as EUROMOD as a starting point. This is a well-established and widely accepted tool, requiring fewer resources for rapid implementation. What’s more, this simplified approach can be seen as an essential step towards identifying avenues that could be of interest and further explored at a later date in terms of expected effects, also considering, on top of behavioral returns during a first period, longer-term perspectives (as in **Liégeois and Genevois, 2015; Liégeois, 2025b**), whereas less sophisticated behavioral contents for the latter. Accordingly, the present paper can be interpreted as a first step in a broader analytical process, paving the way for more in-depth investigations to be carried out at a later stage in support of the social partners, if desired.

The second question relates to concerns about the feasibility of these reforms in political terms. In this sense, and without even mentioning the possible feedback effects of these alternatives, a lighter reform (more progressive, acceptable thresholds or rates, political agenda, etc.) could be considered preferable although less promising in terms of total revenues or impact on inequalities.

All this could open up tracks for future developments and questioning, if one of the alternatives examined were considered sufficiently ripe for general debate, in line with a specific concern and/or close to a publicly defined (political) agenda. In this respect, it is noteworthy that the 2026 reform in Luxembourg introduces –among other aspects– a 0.5% increase in pension social contributions for both employees and employers, alongside a similar 0.5% government contribution. It therefore lies between P4 and the *status quo* (STD). This may reflect both a grounded assessment of needs and a cautious political approach. The present analysis may also lead to further work and possible adjustments on related issues, including inequalities and the poverty rate among residents (see **Figures 7 and 8**).

ORCID iD

Philippe Liégeois  <https://orcid.org/0000-0003-2329-5609>

Acknowledgements

The research presented in this paper is part of a study initiated and financed by the Chambre des Salariés du Luxembourg (CSL). Numerous exchanges with experts (social partners) enabled us to provide an approach more in line with the questions and final expectations of our usual public targets. Yet the present paper mentioning Philippe Liégeois as sole author, its outcomes result from a larger network. The developments presented here are initially based on the model **EUROMOD/SILC** version I4.62+ Beta release (3.4.10). Originally maintained, developed and managed by the Institute for Social and Economic Research (ISER), since 2021 EUROMOD is maintained (with regard to resident populations),

23. The RURO approach retains this random utility structure but introduces an additional random opportunity set. The key idea is that observed labour supply choices depend not only on preferences over income and leisure, but also on the set of job opportunities effectively available to the individual. RURO as far as feasible given data constraints is especially relevant when we aim to distinguish more clearly between outcomes driven by preferences and those driven by labour market opportunities. See (**Aaberge et al., 1995; Aaberge et al., 1999**) –discussed and operationalised in **Capéau et al. (2016), Aaberge and Colombino (2018)** or, along the same lines, **Dagsvik and Strøm (2006)**.

developed and managed by the Joint Research Centre (JRC) of the European Commission in Seville, in collaboration with EUROSTAT and national teams from the EU countries (including LISER for Luxembourg). The EUROMOD/HFCS-R model is building on the same core version as EUROMOD/SILC, yet running on HFCS data rather, and was developed for its base versions (socio-fiscal policies as in 2017) by Jonas Boone, Johannes Derboven, Sarah Kuypers and Gerlinde Verbist, from the University of Antwerp, together with Francesco Figari, from the Università degli studi del Piemonte orientale. An extension to the new EUROMOD/HFCS-XB model, involving microdata related to cross-borders for Luxembourg, has been set up by Johannes Derboven, in collaboration with Philippe Liégeois. A specific documentation for extending EUROMOD/HFCS-XB to all cross-borders, through macro adjustments, was also gathered by Anasse El Maslohi, from LISER. Even if HFCS results from a European-wide effort, Michael Ziegelmeyer from the Banque Centrale du Luxembourg, and Carla Martins from LISER, more specifically but among many others, have indirectly supported the present study through their expertise in those data for Luxembourg.

We are also grateful to the editor of the International Journal of Microsimulation and two anonymous referees for their stimulating suggestions. Obviously and meanwhile, the results developed here and their interpretation are the author's sole responsibility at this stage.

Funding

This research was carried out as part of the project "Alternative Ways for Funding the Luxembourgish Social Security System, with Distributional Effects (2022-2023)" funded by the Chambre des Salariés du Luxembourg/CSL under agreement dated 18 July 2022.

Conflict of Interest

No competing interests reported.

Data and code availability

The analysis builds on data from [i] the "European Union Statistics on Income and Living Conditions" (EU-SILC) survey Wave 16, available for scientific research upon request to EUROSTAT, [ii] the third wave of the Eurosystem Household Finance and Consumption Survey (HFCS) for residents, available for scientific research upon request to the European Central Bank, [iii] the third wave of the Luxembourg Household Finance and Consumption Survey for Cross-Border Commuters (for access requests, contact the Luxembourg Institute of Socio-Economic Research at dataservice@liser.lu or marc.schneider@liser.lu) and [iv] the EU-wide micro-simulation model EUROMOD (I4.62+ Beta release / 3.4.10) available upon request to the EU-Joint Research Center in Seville.

Details of STATA routines for the creation of the EUROMOD input datasets from HFCS raw data and specific policy codes are available from the author upon request.

References

- Household Finance and Consumption Survey: Methodological report for the 2017 wave. 2020. *ECB Statistics Paper*, N°. Household Finance and Consumption Network. DOI: <https://doi.org/10.2866/66933>
- Aaberge R, Dagsvik JK, Strøm S, Strøm S. 1995. Labor supply responses and welfare effects of tax reforms. *The Scandinavian Journal of Economics* **97**: 635–659. DOI: <https://doi.org/10.2307/3440547>
- Aaberge R, Colombino U, Strøm S. 1999. Labour supply in Italy: an empirical analysis of joint household decisions, with taxes and quantity constraints. *Journal of Applied Econometrics* **14**: 403–422. DOI: [https://doi.org/10.1002/\(SICI\)1099-1255\(199907/08\)14:4<403::AID-JAE520>3.0.CO;2-4](https://doi.org/10.1002/(SICI)1099-1255(199907/08)14:4<403::AID-JAE520>3.0.CO;2-4)
- Aaberge R, Colombino U. 2018. Structural labour supply models and microsimulation. *International Journal of Microsimulation* **11**: 162–197. DOI: <https://doi.org/10.34196/ijm.00177>
- Berger F, Islam N, Liégeois P. 2011. Behavioural Microsimulation and Female Labour Supply in Luxembourg. *Brussels Economic Review*. Vol. **54** Université Libre de Bruxelles. p. 389–420.
- Blond-Hanten C, Thomas A. 2014. Stakeholders and expertise: obstacles, expectations and dissemination. *Work Package D of the Progress Project MiDLAS*. Vol. 30 LISER.
- Capéau B, Decoster A, Dekkers G. 2016. Estimating and simulating with a random utility random opportunity model of job choice presentation and application to Belgium. *International Journal of Microsimulation* **9**: 144–191. DOI: <https://doi.org/10.34196/ijm.00139>
- Chen Y, Mathä T, Pulina G, Schuster B, Ziegelmeyer M. 2020. The Luxembourg Household Finance and Consumption Survey: Results from the third wave. *BCL Working Papers*. Central Bank of Luxembourg. p. 142.
- Chen Y, Mathä T, Pulina G, Ziegelmeyer M. 2021. The Cross-Border Household Finance and Consumption Survey: Results from the third wave. *BCL Working Papers*. Central Bank of Luxembourg. p. 154.

- Clément F**, Belkacem R, Pigeron-Piroth I, Wille C (Eds). 2023. *Le Travail Frontalier En Europe / Cross-Border Work in Europe - Réalités et Défis / Realities and Challenges*. Larcier - Luxembourg.
- Dagsvik JK**, Strøm S. 2006. Sectoral labour supply, choice restrictions and functional form. *Journal of Applied Econometrics* **21**: 803–826. DOI: <https://doi.org/10.1002/jae.866>
- Dekkers G**, Van den Bosch K, Barslund M, Kirn T, Baumann N, Kump N, Liégeois P, Moreira A, Stropnik N. 2022. How do gendered labour market trends and the pay gap translate into the projected gender pension gap? A comparative analysis of five countries with low, middle and high GPGs. *Social Sciences* **11**: 304. DOI: <https://doi.org/10.3390/socsci11070304>
- European Commission – DG ECFIN**. 2024. 2024 Ageing Report - Economic and Budgetary Projections for the EU Member States (2022-2070). https://economy-finance.ec.europa.eu/publications/2024-ageing-report-economic-and-budgetary-projections-eu-member-states-2022-2070_en
- Greco S**, Ishizaka A, Tasiou M, Torrisi G. 2019. On the methodological framework of composite indices: A review of the issues of weighting, aggregation, and robustness. *Social Indicators Research* **141**: 61–94. DOI: <https://doi.org/10.1007/s11205-017-1832-9>
- Islam N**, El Maslohi A, A-So G. 2020. EUROMOD Country Report for Luxembourg, 2017-2020 [LISER and University of Essex]. <https://euromod-web.jrc.ec.europa.eu/resources/country-reports/f3-onwards/l30plus>
- Kuypers S**, Figari F, Verbist G, Verbist G. 2016. The eurosystem household finance and consumption survey: A new underlying database for EUROMOD. *International Journal of Microsimulation* **9**: 35–65. DOI: <https://doi.org/10.34196/ijm.00142>
- Kuypers S**, Boone J, Derboven J, Figari F, Verbist G. 2020. Enhancing microsimulation analysis of wealth-related policies in EUROMOD. *International Journal of Microsimulation* **13**: 5–26. DOI: <https://doi.org/10.34196/IJM.00223>
- Liégeois PH**, Genevois A-S. 2015. LuDMi – Dynamic Microsimulation Model for Luxembourg”, Technical report. EU-PROGRESS MiDLAS Project (2013-2015). p. 81.
- Liégeois P**. 2023a. *Alternative Ways for Funding the Luxembourgish Social Security System, with Distributional Effects - Final Report Involving Both LU-Residents and Cross-Border Households Analyzed Using Static EUROMOD HFCS-Based Models*. LISER and CSL, Mimeo, Confidential.
- Liégeois PH**. 2023b. *Financement Du Système de Sécurité Sociale Luxembourgeois - Analyse Statique Exploratoire de Pistes Alternatives, Avec Effets Redistributifs, Impliquant à La Fois Les Ménages Résidents et Frontaliers – Modèles EUROMOD/HFCS - Synthèse Pour Un Public Élargi*. LISER, CSL and partners, mimeo (confidential).
- Liégeois P**. 2025a. Dealing with Residents and Cross-borders in a Tax-benefit Model for Luxembourg. *International Journal of Microsimulation* **18**. DOI: <https://doi.org/10.34196/ijm.00324>
- Liégeois P**. 2025b. Addressing the Social Sustainability of Pensions in Luxembourg through Dynamic Microsimulation. LISER, Technical Report. Mimeo. p. 22.
- Mazziotta M**, Pareto A. 2011. Methods for constructing composite indices: One for all or all for one. *Italian Journal of Economic, Demographic and Statistical Studies* **67**: 67–80.
- Mazziotta M**, Pareto A. 2016. On a Generalized Non-compensatory Composite Index for Measuring Socio-economic Phenomena. *Social Indicators Research* **127**: 983–1003. DOI: <https://doi.org/10.1007/s11205-015-0998-2>
- Nardo M**, Saisana M, Saltelli A, Tarantola S, Hoffman A, Giovannini E. 2008. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD & European Commission, Joint Research Centre.
- Ochmann R**, Blömer M, Haan P, Müller K-U, Tomasch E. 2014. “Studie zu alternativen Finanzierungsansätzen für die Sozialversicherungssysteme Luxemburgs”, Deutsches Institut für Wirtschaftsforschung (DIW Berlin). https://www.csl.lu/app/uploads/2021/02/2014-04-01_diw-finanzierungsansatze-2.-endbericht.pdf
- OECD**. 2023. OECD statistics (as on early 2023). <http://stats.oecd.org/Index.aspx>
- Sologon D**, O'Donoghue C, Andreoli F, Gerber P, Licheron J, Montes Vinas A, Paccoud A. 2023. Spatial Economics of Income Distribution Across Borders: Drivers of Spatial Inequalities using Microsimulation. LISER (on-going project). <https://liser.elsevierpure.com/fr/projects/spatial-economics-of-income-distribution-across-borders-drivers-o>
- STATEC**. 2019. *Le Luxembourg En Chiffres*. p. 53.
- STATEC**. 2023. *Le Luxembourg en chiffres*. <https://statistiques.public.lu/fr/publications/series/luxembourg-en-chiffres/2023/luxembourg-en-chiffres-2023.html>
- Sutherland H**, Figari F. 2013. EUROMOD: the European Union tax-benefit microsimulation model. *International Journal of Microsimulation* **6**: 4–26. DOI: <https://doi.org/10.34196/ijm.00075>
- Vergnat V**, D'Ambrosio C, Liégeois P. 2022. The distributive impact of the Luxembourg tax-benefit system: a more comprehensive measurement. *Public Finance Review* **50**: 436–483. DOI: <https://doi.org/10.1177/10911421221113842>

Appendix A

Positioning the Indicators Retained for a Synthetic View of Alternatives Within a Broader Family of possibilities

Each alternative might be described through an arbitrary series of basic indicators of interest, indeed here -and as an exploratory track only- ten of them initially chosen on an *ad hoc* way grounding on best practices. Therefore, an initial step for reducing the complexity of a general comparative analysis might be to look for a subset of indicators that would represent at best each alternative.

Table A1. Correlations between an *ad hoc* selection of basic outcome indicators

Correlations	Total Receipts	Receipts from Residents	Receipts from XBs	GINI Residents	POVERTY Residents	POVERTY - Single Resident Households with Dependents	WELL-BEING Residents	GINI XBs	POVERTY XBs	WELL-BEING XBs
Total Receipts	1.00									
Receipts from Residents	1.00	1.00								
Receipts from XBs	0.99	0.97	1.00							
GINI Residents	-0.40	-0.45	-0.26	1.00						
POVERTY Residents	-0.11	-0.14	-0.05	-0.01	1.00					
POVERTY - Single Resident Households with Dependents	-0.21	-0.23	-0.16	-0.08	0.82	1.00				
WELL-BEING Residents	-0.97	-0.98	-0.93	0.49	0.23	0.33	1.00			
GINI XBs	-0.42	-0.44	-0.35	0.68	0.19	-0.07	0.44	1.00		
POVERTY XBs	-0.81	-0.79	-0.82	0.10	0.41	0.47	0.81	0.38	1.00	
WELL-BEING XBs	-0.98	-0.97	-0.98	0.28	0.16	0.30	0.96	0.35	0.87	1.00

Note to the reader: the correlations refer to the outcomes for all (43) alternatives ; indicators referring to XBs are shown in blue fonts (these refer to the populations covered by the HFCS-XB survey, out of the (total) Receipts from XBs which are macro-adjusted for "non-active" XB households).

For that purpose, we derive in **Table A1** the correlations between those classical indicators. These are correlations only, signaling that a couple of dimensions are more or less varying throughout the alternatives in a regular way, whatever the reason why, and definitively not an indication of *causal* relationships. A strong correlation (close to extremes "1" or "-1") may also result from a common other determinant (latent variable). For example, strongly increasing the taxes on income deciles far above the median while reducing them around the median might reduce overall inequalities (Gini index lowered) while leading to some higher poverty rate (the median income, hence the well-being, being pushed upward whereas lower deciles are not affected in absolute terms by the reform), which does not imply that a policy reducing inequalities is genuinely increasing the poverty.

Our objective is to build our comparative analysis on a maximum of three basic indicators for each alternative, for more efficient graphical representations and classifications, hence easier interpretations. In this sense, we can conclude from **Table A** that the total receipts (for both residents and XBs), the Gini index for residents and the poverty rate for residents might become, among other possibilities but in line with best practices anyway, such a relevant triple.

On the one side, none of those indicators are strongly correlated to the other two, hence each of them providing to a certain extent some additional information by itself.

On the other side, the poverty of residents seems strongly and positively correlated to the poverty of single resident households with dependents (+) and the total receipts more correlated to the receipts from XBs or residents (+), the average well-beings (-) and the poverty of XBs (-). The Gini coefficient of residents is also somewhat in line with the one for XBs (+). Therefore and building on the triple of basic indicators retained, we might cautiously derive some underlying qualitative priors about the possible effect of an alternative on other dimensions.

Of course, such a basic approach to indicator selection is questionable, and other choices could have been made, either in terms of the indicators selected (for example, poverty among single parent households might be another relevant candidate), or in terms of the technology chosen for a selection of them. But we are mindful of the need for the indicators presented to our partners to be readable and familiar enough.

Appendix B

Towards a Basic Global Performance Index

Table B1 is showing up the first and second best alternative systems of policies, as well as the worst one (in grey font), identified through their global performance index GPI defined in Section 4.2, taking into account several values for the weights a (Gini coefficient, vertically), β (poverty rate, horizontally) and g (total receipts, not explicit here as simply equal to $1-a-\beta$). As a matter of references, **Table B1** is also reminding on its top left corner the overall minimum and maximum values for the three selected basic indicators.

For example, the triple of weights ($a=0$ /Gini, $b=0$ /poverty, $g=1$ /receipts), exclusively concentrating in total receipts for Luxembourg (from both residents and XBs), is leading to C3 as the best possible system (12,432 million €/year for the total receipts), in conformity with the maximum identified in Graph 3. But C4 becomes the best system if the Gini coefficient is at stake exclusively ($a=1$ /Gini, $b=0$ /poverty, $g=0$ /receipts): 0.2922 at best. C4 remains the best system when the triple ($a=0$ /Gini, $b=1$ /poverty, $g=0$ /receipts) is chosen rather, hence designating the poverty rate as a single target (12.6%). We might also look for a more balanced alternative, for example with weights ($a=0.3$ /Gini, $b=0.3$ /poverty, $g=0.4$ /receipts), which would emphasize the system C4 again. All those values can be compared in Table B-1 with the ones prevailing in the STD system for Luxembourg in 2017 (left side of **Table B1**, reminding **Table 1**).

The combined systems C1-C4 playing a specific and prominent role in terms of total receipts (they have been designed with such a criteria in mind, leaving aside their possibly important 2nd round effects), **Table B2**, structured in the same way as **Table B1**, is showing up the best systems of policies when C1-C4 are excluded rather. In such a framework, I3 becomes the best system when the Gini is at stake exclusively (0.2939), PE if the poverty rate is emphasized¹ (12.6%), PC for the total receipts (11,558 million €/year) and PF -as evoked already in Section 4.2-for a more balanced alternative.

Table B1. Global performance index, given weights for Gini coefficient, poverty rate and total receipts (in million € / year) All alternatives considered (including the combined systems C1-C4)

p.m. RECEIPTS (in millions €/year) (Min: 9,849 - Max : 12,432)

GINI (Min: 0.2922 - Max : 0.3004) a β	POVERTY / b β (Min: 12.6% - Max: 14.7%)													
	0			0.3			0.6			1				
Best Systems, "Cx" INCLUDED														
	INDIC	STD	C3	C4	C2	C3	C4	C2	C4	C2	C1	C4	PE	HA
0	REC	10,185	12,432	11,670	11,655	12,432	11,670	11,655	11,670	11,655	11,557	11,670	10,361	10,488
	GINI_R	0.2993	0.2971	0.2922	0.2927	0.2971	0.2922	0.2927	0.2922	0.2927	0.2935	0.2922	0.2985	0.2964
	POV_R	13.2%	13.6%	12.6%	13.0%	13.6%	12.6%	13.0%	12.6%	13.0%	13.0%	12.6%	12.6%	12.7%
	INDIC	STD	C3	C4	C2	C4	C2	C1	C4	C2	C1			
	REC	10,185	12,432	11,670	11,655	11,670	11,655	11,557	11,670	11,655	11,557			
0.3	GINI_R	0.2993	0.2971	0.2922	0.2927	0.2922	0.2927	0.2935	0.2922	0.2927	0.2935			
	POV_R	13.2%	13.6%	12.6%	13.0%	12.6%	13.0%	13.0%	12.6%	13.0%	13.0%			
	INDIC	STD	C4	C2	C1	C4	C2	C1						
	REC	10,185	11,670	11,655	11,557	11,670	11,655	11,557						
	GINI_R	0.2993	0.2922	0.2927	0.2935	0.2922	0.2927	0.2935						
0.6	POV_R	13.2%	12.6%	13.0%	13.0%	12.6%	13.0%	13.0%						

1. PE was also 2nd best with that respect when alternatives "Cx" were included (see **Table B1**).

p.m. RECEIPTS (in millions €/year) (Min: 9,849 - Max : 12,432)

1	INDIC	STD	C4	C2	C1
	REC	10,185	11,670	11,655	11,557
	GINI_R	0.2993	0.2922	0.2927	0.2935
	POV_R	13.2%	12.6%	13.0%	13.0%

Table B2. Global Performance Index, given weights for Gini coefficient, poverty rate and total Receipts (in million € / year) - All alternatives (excluding the combined systems C1-C4)

p.m. RECEIPTS (in millions €/year) (Min: 9,849 - Max : 11,558)

GINI (Min: 0.2939 - Max : 0.3004) a b	POVERTY / b (Min: 12.6% - Max: 14.7%)		0			0.3			0.6			1		
	Best Systems, "CX" EXCLUDED		1st Best	2nd Best	3rd Best	1st Best	2nd Best	3rd Best	1st Best	2nd Best	3rd Best	1st Best	2nd Best	3rd Best
	INDIC	STD	PC	PF	PB	PC	PF	PB	PF	HA	PE	PE	HA	PF
	REC	10,185	11,558	10,793	10,782	11,558	10,793	10,782	10,793	10,488	10,361	10,361	10,488	10,793
	GINI_R	0.2993	0.3004	0.2957	0.2960	0.3004	0.2957	0.2960	0.2957	0.2964	0.2985	0.2985	0.2964	0.2957
0	POV_R	13.2%	13.7%	12.8%	13.2%	13.7%	12.8%	13.2%	12.8%	12.7%	12.6%	12.6%	12.7%	12.8%
	INDIC	STD	PC	PF	PB	PF	HA	PB	HA	PF	PE			
	REC	10,185	11,558	10,793	10,782	10,793	10,488	10,782	10,488	10,793	10,361			
	GINI_R	0.2993	0.3004	0.2957	0.2960	0.2957	0.2964	0.2960	0.2964	0.2957	0.2985			
0.3	POV_R	13.2%	13.7%	12.8%	13.2%	12.8%	12.7%	13.2%	12.7%	12.8%	12.6%			
	INDIC	STD	I3	PF	PB	PF	HA	PB						
	REC	10,185	10,221	10,793	10,782	10,793	10,488	10,782						
	GINI_R	0.2993	0.2939	0.2957	0.2960	0.2957	0.2964	0.2960						
0.6	POV_R	13.2%	14.7%	12.8%	13.2%	12.8%	12.7%	13.2%						
1	INDIC	STD	I3	I6	PF									
	REC	10,185	10,221	9,849	10,793									
	GINI_R	0.2993	0.2939	0.2945	0.2957									
	POV_R	13.2%	14.7%	14.0%	12.8%									